

```
mysql> use proj;
```

Database changed

Create and Insert

```
mysql> CREATE TABLE departments (
```

```
-> dept_id INT PRIMARY KEY,  
-> dept_name VARCHAR(50),  
-> location VARCHAR(50)  
-> );
```

```
mysql> INSERT INTO departments VALUES
```

```
-> (101,'HR','Delhi'),  
-> (102,'IT','Mumbai'),  
-> (103,'Finance','Pune'),  
-> (104,'Sales','Bangalore'),  
-> (105,'Marketing','Hyderabad');
```

```
mysql> CREATE TABLE employees (
```

```
-> emp_id INT PRIMARY KEY,  
-> emp_name VARCHAR(50),  
-> gender VARCHAR(10),  
-> age INT,  
-> salary DECIMAL(10,2),  
-> joining_date DATE,  
-> dept_id INT,  
-> manager_id INT,  
-> FOREIGN KEY (dept_id) REFERENCES departments(dept_id)  
-> );
```

```
mysql> INSERT INTO employees VALUES
```

```
-> (1,'Amit','Male',28,45000,'2021-02-15',102,NULL),  
-> (2,'Neha','Female',25,38000,'2022-01-20',101,1),  
-> (3,'Rahul','Male',32,70000,'2020-08-18',102,1),  
-> (4,'Priya','Female',29,65000,'2019-05-10',103,3),  
-> (5,'Rohan','Male',35,90000,'2018-03-22',104,NULL),  
-> (6,'Sneha','Female',27,42000,'2023-01-12',101,2),  
-> (7,'Arjun','Male',31,80000,'2017-06-05',102,3),  
-> (8,'Pooja','Female',26,39000,'2022-10-10',105,5),  
-> (9,'Vikas','Male',30,55000,'2021-07-01',104,5),  
-> (10,'Kiran','Female',24,36000,'2023-04-14',103,4),  
-> (11,'Ankit','Male',29,52000,'2021-12-11',102,3),  
-> (12,'Meera','Female',33,76000,'2019-09-21',105,5),  
-> (13,'Riya','Female',28,47000,'2022-06-16',101,2),  
-> (14,'Mohit','Male',34,87000,'2018-11-19',104,5),  
-> (15,'Nisha','Female',31,62000,'2020-01-09',103,4);
```

```
mysql> CREATE TABLE projects (  
->  project_id INT PRIMARY KEY,  
->  project_name VARCHAR(100),  
->  start_date DATE,  
->  end_date DATE,  
->  budget DECIMAL(12,2)  
-> );
```

```
mysql> INSERT INTO projects VALUES  
-> (201,'AI Chatbot','2023-01-01','2023-12-31',1000000),  
-> (202,'ERP System','2022-05-01','2023-11-30',800000),  
-> (203,'Website','2023-03-15','2023-09-30',400000),  
-> (204,'Mobile App','2022-10-10','2024-01-31',1200000),  
-> (205,'CRM System','2023-02-20','2023-10-31',600000);
```

Query OK, 5 rows affected (0.086 sec)
Records: 5 Duplicates: 0 Warnings: 0

```
mysql> select * from departments;
```

dept_id	dept_name	location
101	HR	Delhi
102	IT	Mumbai
103	Finance	Pune
104	Sales	Bangalore
105	Marketing	Hyderabad

Select Query

```
mysql> select * from employees;
```

emp_id	emp_name	gender	age	salary	joining_date	dept_id	manager_id
1	Amit	Male	28	45000.00	2021-02-15	102	NULL
2	Neha	Female	25	38000.00	2022-01-20	101	1
3	Rahul	Male	32	70000.00	2020-08-18	102	1
4	Priya	Female	29	65000.00	2019-05-10	103	3
5	Rohan	Male	35	90000.00	2018-03-22	104	NULL
6	Sneha	Female	27	42000.00	2023-01-12	101	2
7	Arjun	Male	31	80000.00	2017-06-05	102	3
8	Pooja	Female	26	39000.00	2022-10-10	105	5
9	Vikas	Male	30	55000.00	2021-07-01	104	5
10	Kiran	Female	24	36000.00	2023-04-14	103	4
11	Ankit	Male	29	52000.00	2021-12-11	102	3

```
| 12 | Meera | Female | 33 | 76000.00 | 2019-09-21 | 105 | 5 |
| 13 | Riya | Female | 28 | 47000.00 | 2022-06-16 | 101 | 2 |
| 14 | Mohit | Male | 34 | 87000.00 | 2018-11-19 | 104 | 5 |
| 15 | Nisha | Female | 31 | 62000.00 | 2020-01-09 | 103 | 4 |
+-----+-----+-----+-----+-----+-----+-----+
```

15 rows in set (0.044 sec)

```
mysql> select * from projects;
```

```
+-----+-----+-----+-----+-----+
| project_id | project_name | start_date | end_date | budget |
+-----+-----+-----+-----+-----+
| 201 | AI Chatbot | 2023-01-01 | 2023-12-31 | 1000000.00 |
| 202 | ERP System | 2022-05-01 | 2023-11-30 | 800000.00 |
| 203 | Website | 2023-03-15 | 2023-09-30 | 400000.00 |
| 204 | Mobile App | 2022-10-10 | 2024-01-31 | 1200000.00 |
| 205 | CRM System | 2023-02-20 | 2023-10-31 | 600000.00 |
+-----+-----+-----+-----+-----+
```

```
mysql> create table manager(
```

```
-> manager_id int primary key,
-> manager_name varchar(100));
```

```
mysql> INSERT INTO manager VALUES
```

```
-> (1, 'John'),
-> (2, 'David'),
-> (3, 'Sarah'),
-> (4, 'Dhanvanti'),
-> (5, 'Umesh');
```

Adding Manager_Id As Foreign Key

```
mysql> ALTER TABLE employees
```

```
-> ADD FOREIGN KEY (manager_id)
-> REFERENCES manager(manager_id);
```

Describe Table

```
mysql> desc employees
```

```
-> ;
```

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
emp_name	varchar(50)	YES		NULL	
gender	varchar(10)	YES		NULL	
age	int	YES		NULL	
salary	decimal(10,2)	YES		NULL	
joining_date	date	YES		NULL	
dept_id	int	YES	MUL	NULL	
manager_id	int	YES	MUL	NULL	

8 rows in set (0.779 sec)

mysql> desc manager;

Field	Type	Null	Key	Default	Extra
manager_id	int	NO	PRI	NULL	
manager_name	varchar(100)	YES		NULL	

Questions :

mysql> -- Employees earning more than 50000 AND age above 30. **(Comment)**

mysql> select emp_id,emp_name,age,salary from employees

-> where salary>=50000 and age>30;

emp_id	emp_name	age	salary
3	Rahul	32	70000.00
5	Rohan	35	90000.00
7	Arjun	31	80000.00
12	Meera	33	76000.00
14	Mohit	34	87000.00
15	Nisha	31	62000.00

mysql> -- Employees from IT OR HR. **(Comment)**

mysql> select e.emp_id,e.emp_name,d.dept_id,d.dept_name from employees as e

-> join departments as d

-> on e.dept_id = d.dept_id

-> where d.dept_name in ('IT','HR');

```
+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | dept_name |
+-----+-----+-----+-----+
| 2 | Neha | 101 | HR |
| 6 | Sneha | 101 | HR |
| 13 | Riya | 101 | HR |
| 1 | Amit | 102 | IT |
| 3 | Rahul | 102 | IT |
| 7 | Arjun | 102 | IT |
| 11 | Ankit | 102 | IT |
+-----+-----+-----+-----+
```

mysql> -- Employees NOT in Sales. **(Comment)**

mysql> select e.emp_id,e.emp_name,d.dept_id,d.dept_name from employees as e

-> join departments as d

-> on e.dept_id = d.dept_id

-> where d.dept_name not in ('Sales');

```
+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | dept_name |
+-----+-----+-----+-----+
| 2 | Neha | 101 | HR |
| 6 | Sneha | 101 | HR |
| 13 | Riya | 101 | HR |
| 1 | Amit | 102 | IT |
| 3 | Rahul | 102 | IT |
| 7 | Arjun | 102 | IT |
| 11 | Ankit | 102 | IT |
| 4 | Priya | 103 | Finance |
| 10 | Kiran | 103 | Finance |
| 15 | Nisha | 103 | Finance |
| 8 | Pooja | 105 | Marketing |
| 12 | Meera | 105 | Marketing |
+-----+-----+-----+-----+
```

mysql> -- Salary BETWEEN 40000 AND 80000. **(Comment)**

mysql> select emp_id,emp_name,salary from employees

-> where salary between 40000 and 80000;

+-----+-----+-----+		
emp_id	emp_name	salary
+-----+-----+-----+		
1	Amit	45000.00
3	Rahul	70000.00
4	Priya	65000.00
6	Sneha	42000.00
7	Arjun	80000.00
9	Vikas	55000.00
11	Ankit	52000.00
12	Meera	76000.00
13	Riya	47000.00
15	Nisha	62000.00
+-----+-----+-----+		

mysql> -- Name LIKE 'A%' **(Comment)**

mysql> select emp_name from employees
-> where emp_name like 'A%';

+-----+	
emp_name	
+-----+	
Amit	
Arjun	
Ankit	
+-----+	

mysql> -- Name LIKE '%a' **(Comment)**

mysql> select emp_name from employees
-> where emp_name like '%a';

+-----+	
emp_name	
+-----+	
Neha	
Priya	
Sneha	
Pooja	
Meera	
Riya	
Nisha	
+-----+	

mysql> -- Name LIKE '%ra%' **(Comment)**

mysql> select emp_name from employees
-> where emp_name like '%ra%';

emp_name
Rahul
Kiran
Meera

mysql> SELECT emp_name
-> FROM employees
-> WHERE LOWER(emp_name) LIKE '%ra%';

emp_name
Rahul
Kiran
Meera

3 rows in set (0.006 sec)

mysql> -- salary is not equal to 50000 **(Comment)**
Query OK, 0 rows affected (0.004 sec)

mysql> select emp_name,salary from employees
-> where salary <> 50000;

emp_name	salary
Amit	45000.00
Neha	38000.00
Rahul	70000.00
Priya	65000.00
Rohan	90000.00
Sneha	42000.00
Arjun	80000.00
Pooja	39000.00
Vikas	55000.00
Kiran	36000.00

```
| Ankit | 52000.00 |
| Meera | 76000.00 |
| Riya | 47000.00 |
| Mohit | 87000.00 |
| Nisha | 62000.00 |
```

```
+-----+-----+
```

```
mysql> select emp_id,emp_name,salary from employees
```

```
-> where salary <> 80000;
```

```
+-----+-----+-----+
```

```
| emp_id | emp_name | salary |
```

```
+-----+-----+-----+
```

```
| 1 | Amit | 45000.00 |
| 2 | Neha | 38000.00 |
| 3 | Rahul | 70000.00 |
| 4 | Priya | 65000.00 |
| 5 | Rohan | 90000.00 |
| 6 | Sneha | 42000.00 |
| 8 | Pooja | 39000.00 |
| 9 | Vikas | 55000.00 |
| 10 | Kiran | 36000.00 |
| 11 | Ankit | 52000.00 |
| 12 | Meera | 76000.00 |
| 13 | Riya | 47000.00 |
| 14 | Mohit | 87000.00 |
| 15 | Nisha | 62000.00 |
```

```
+-----+-----+-----+
```

```
mysql> -- Manager IS NULL and Manager IS not NULL. (Comment)
```

```
mysql> select emp_id,emp_name,manager_id from employees
```

```
-> where manager_id is null;
```

```
+-----+-----+-----+
```

```
| emp_id | emp_name | manager_id |
```

```
+-----+-----+-----+
```

```
| 1 | Amit | NULL |
| 5 | Rohan | NULL |
```

```
+-----+-----+-----+
```

```
2 rows in set (0.031 sec)
```

```
mysql> select emp_id,emp_name,manager_id from employees
```

```
-> where manager_id is not null;
```

```
+-----+-----+-----+
```

```
| emp_id | emp_name | manager_id |
```

```
+-----+-----+-----+
```


	2	Neha		1	
	3	Rahul		1	
	4	Priya		3	
	6	Sneha		2	
	7	Arjun		3	
	8	Pooja		5	
	9	Vikas		5	
	10	Kiran		4	
	11	Ankit		3	
	12	Meera		5	
	13	Riya		2	
	14	Mohit		5	
	15	Nisha		4	

```

+-----+-----+-----+

```

mysql> -- Employees satisfying multiple conditions using parentheses. **(Comment)**

```

mysql> SELECT emp_id, emp_name, age, salary, dept_id
    -> FROM employees
    -> WHERE (age > 30 AND salary > 50000)
    -> OR dept_id = 2;

```

```

+-----+-----+-----+-----+-----+
| emp_id | emp_name | age | salary | dept_id |
+-----+-----+-----+-----+-----+
| 3 | Rahul | 32 | 70000.00 | 102 |
| 5 | Rohan | 35 | 90000.00 | 104 |
| 7 | Arjun | 31 | 80000.00 | 102 |
| 12 | Meera | 33 | 76000.00 | 105 |
| 14 | Mohit | 34 | 87000.00 | 104 |
| 15 | Nisha | 31 | 62000.00 | 103 |
+-----+-----+-----+-----+-----+

```

mysql> -- Employees from Mumbai location **(Comment)**

Employees from Mumbai location. **(Comment)**

```

mysql> SELECT e.emp_id,e.emp_name,e.age,e.salary, d.dept_id,d.location from employees as e
    -> join departments as d
    -> on e.dept_id = d.dept_id
    -> where d.location = 'Mumbai';

```

```

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | age | salary | dept_id | location |
+-----+-----+-----+-----+-----+-----+

```

	1	Amit		28	45000.00		102	Mumbai	
	3	Rahul		32	70000.00		102	Mumbai	
	7	Arjun		31	80000.00		102	Mumbai	
	11	Ankit		29	52000.00		102	Mumbai	

+-----+-----+-----+-----+-----+

Employee - Joined after 2021 **(Comment)**

select emp_id,emp_name,joining_date from employees
-> where joining_date > '2021-02-15';

	emp_id		emp_name		joining_date	
--	--------	--	----------	--	--------------	--

+-----+-----+-----+

	2	Neha		2022-01-20	
	6	Sneha		2023-01-12	
	8	Pooja		2022-10-10	
	9	Vikas		2021-07-01	
	10	Kiran		2023-04-14	
	11	Ankit		2021-12-11	
	13	Riya		2022-06-16	

+-----+-----+-----+

Joined before 2020 **(Comment)**

-> select emp_id,emp_name,joining_date from employees
-> where joining_date < '2020-01-09';

	emp_id		emp_name		joining_date	
--	--------	--	----------	--	--------------	--

+-----+-----+-----+

	4	Priya		2019-05-10	
	5	Rohan		2018-03-22	
	7	Arjun		2017-06-05	
	12	Meera		2019-09-21	
	14	Mohit		2018-11-19	

+-----+-----+-----+